

D904

Algebra for Secondary Mathematics Teaching

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

For a department meeting on the third Thursday of the month, a teacher sorts three examples: $3x + 5 = 11$, $a + b = b + a$, and $y = 2x + 3$. The meeting room is next to the library. In which example does the letter act as a generalized number?

- A. $y = 2x + 3$
- B. $3x + 5 = 11$
- C. The letter plays the same role in all three.
- D. $a + b = b + a$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A teacher preparing a warm-up for a class of 28 wants students to use structure to solve $(x - 3)^2 = 16$ without expanding. The warm-up will be projected on a new interactive screen. What are the solutions?

- A. $x = 19$
- B. $x = -7$ or $x = 1$
- C. $x = 7$ or $x = -1$
- D. $x = 7$ only

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A theater that was renovated in 2019 has 22 seats in the first row, and each row has 3 more seats than the row in front of it. Tickets cost \$18. How many seats are in row 15?

- A. $22 \cdot 3^{14}$
- B. 67
- C. 64
- D. 45

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

An arithmetic sequence has $a_1 = 5$ and $a_{10} = 50$. What is the common difference?

- A. 4.5
- B. 5
- C. 10
- D. 45

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

On a windless afternoon, a ball weighing 450 grams is thrown upward; its height is $h(t) = -5t^2 + 30t$ meters. What is its average rate of change in height from $t = 1$ to $t = 3$ seconds?

- A. 10 meters per second
- B. 20 meters per second
- C. 15 meters per second
- D. -10 meters per second

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A function has an average rate of change of 6 on the interval $[2, 5]$, and $f(2) = 4$. What is $f(5)$?

- A. 30
- B. 22
- C. -14
- D. 10

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A teacher shares a translated 9th-century problem written entirely in words: “a square and ten roots equal thirty-nine,” solved by completing a drawn square. She found the translation in a museum gift shop. Which stage of algebra does the problem show?

- A. Symbolic algebra.
- B. Not algebra at all, because it uses no letters.
- C. Rhetorical algebra.
- D. Syncopated algebra.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A teacher models $(x + 3)(x + 2)$ with algebra tiles from a plastic tub of 200 pieces. The lesson falls in the second week of the semester. What total area do the tiles show?

- A. $x^2 + 6$
- B. $x^2 + 6x + 5$
- C. $x^2 + 5x + 5$
- D. $x^2 + 5x + 6$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

An area model has total area $x^2 + 7x + 12$. What are its dimensions?

- A. $(x + 7)$ and $(x + 12)$
- B. $(x + 3)$ and $(x + 4)$
- C. $(x + 6)$ and $(x + 2)$
- D. $(x - 3)$ and $(x - 4)$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A teacher writing a quiz on a laptop with a cracked screen defines $f(x) = 2x + 1$ and $g(x) = x^2$. The quiz has 10 questions. What is $f(g(3))$?

- A. 19
- B. 16
- C. 63
- D. 49

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

$h(x) = (x + 4)^2$ is written as $f(g(x))$ with $f(x) = x^2$. What is $g(x)$?

- A. $g(x) = x^2 + 4$
- B. $g(x) = x - 4$
- C. $g(x) = (x + 4)^2$
- D. $g(x) = x + 4$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

On a Tuesday quiz with 12 items, a student writes $2(x + 5) = 2x + 5$ on three problems. The student's other answers were neat and on time. What should the teacher do next?

- A.** Name the incomplete-distribution error and test it with numbers or an area model, such as $2(3 + 5) = 16$, not 11.
- B.** Reteach basic multiplication facts.
- C.** Assign 20 more distribution problems of the same type.
- D.** Conclude the student is not ready for algebra.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

For a department meeting ~~on the third Thursday of the month~~, a teacher sorts three examples: $3x + 5 = 11$, $a + b = b + a$, and $y = 2x + 3$. ~~The meeting room is next to the library.~~ In which example does the letter act as a generalized number?

Target word: role of the letter **Decides:** An identity is true for every value of its letters.

A	Adjacent category	Here x and y are quantities that vary together.
B	Confusion set	Here x is a specific unknown, 2.
C	Absolute language	The roles differ; that is the point of the sort.
D	Correct	The statement holds for all numbers; the letters stand for any number.

2. Answer C Concept 2

A teacher preparing a warm-up ~~for a class of 28~~ wants students to use structure to solve $(x - 3)^2 = 16$ without expanding. ~~The warm up will be projected on a new interactive screen.~~ What are the solutions?

Target word: solutions using structure **Decides:** $x - 3 = \pm 4$.

A	Confusion set	Treats $(x - 3)^2$ as $x - 3$, ignoring the square.
B	Inverted operation	Subtracts 3 instead of adding it back.
C	Correct	$x - 3 = 4$ gives 7; $x - 3 = -4$ gives -1 .
D	Dropped element	Drops the negative square root.

3. Answer C Concept 3

A theater ~~that was renovated in 2019~~ has 22 seats in the first row, and each row has 3 more seats than the row in front of it. ~~Tickets cost \$18.~~ How many seats are in row 15?

Target word: 15th term of an arithmetic sequence **Decides:** $a_{15} = 22 + (15 - 1) \cdot 3$.

A	Adjacent category	A geometric formula for a pattern that adds, not multiplies.
B	Wrong stem number	Uses 15 steps instead of 14; row 1 has no added seats.
C	Correct	$22 + 14 \times 3 = 64$.
D	Dropped element	15×3 leaves out the 22 seats in row 1.

4. Answer B Concept 3 • transfer

An arithmetic sequence has $a_1 = 5$ and $a_{10} = 50$. What is the common difference?

Target word: solve for d **Decides:** $50 = 5 + 9d$.

A	Wrong stem number	Divides by 10 instead of the 9 steps between terms.
B	Correct	$9d = 45$, so $d = 5$.
C	Adjacent category	$50 \div 5$ is a ratio, the geometric idea.
D	Dropped element	The total change, not divided among the steps.

5. Answer A Concept 4

~~On a windless afternoon, a ball weighing 450 grams is thrown upward; its height is $h(t) = -5t^2 + 30t$ meters. What is its average rate of change in height from $t = 1$ to $t = 3$ seconds?~~

Target word: average rate of change **Decides:** $h(1) = 25$ and $h(3) = 45$.

A	Correct	$(45 - 25) / (3 - 1) = 10$.
B	Dropped element	The change in height, not divided by the 2 seconds.
C	Confusion set	$45 \div 3$ averages from $t = 0$, not over $[1, 3]$.
D	Inverted operation	Subtracts in the wrong order in the numerator only.

6. Answer B Concept 4 • transfer

A function has an average rate of change of 6 on the interval $[2, 5]$, and $f(2) = 4$. What is $f(5)$?

Target word: solve for an output **Decides:** $f(5) - 4 = 6 \times 3$.

A	Wrong stem number	6×5 uses the endpoint, not the width.
B	Correct	Change in output $= 6 \times 3 = 18$; $4 + 18 = 22$.
C	Inverted operation	Subtracts the change instead of adding it.
D	Dropped element	Adds 6 once, forgetting the interval is 3 units wide.

7. Answer C Concept 5

A teacher shares a translated 9th-century problem written entirely in words: “a square and ten roots equal thirty-nine,” solved by completing a drawn square. ~~She found the translation in a museum gift shop.~~ Which stage of algebra does the problem show?

Target word: historical stage **Decides:** The problem uses words only, no symbols or abbreviations.

A	Adjacent category	Symbolic algebra writes $x^2 + 10x = 39$ with letters and signs.
B	Absolute language	Solving for an unknown quantity is algebra in any notation.
C	Correct	Everything, including the unknown, is stated in words.
D	Confusion set	Syncopated algebra uses some abbreviations; this has none.

8. Answer D Concept 6

A teacher models $(x + 3)(x + 2)$ with algebra tiles ~~from a plastic tub of 200 pieces. The lesson falls in the second week of the semester.~~ What total area do the tiles show?

Target word: product from the area model **Decides:** One x^2 , $3x + 2x = 5x$, and $3 \times 2 = 6$ unit tiles.

A	Dropped element	Leaves out the two middle rectangles.
B	Inverted operation	Multiplies 3 and 2 for the middle and adds them for the constant.
C	Confusion set	Uses the sum of 3 and 2 for both terms.
D	Correct	The rectangle holds x^2 , five x -tiles, and six unit tiles.

9. Answer B Concept 6 • transfer

An area model has total area $x^2 + 7x + 12$. What are its dimensions?

Target word: factor from the area **Decides:** Two numbers with product 12 and sum 7.

A	Wrong stem number	Copies the coefficients as the dimensions.
B	Correct	$3 \times 4 = 12$ and $3 + 4 = 7$.
C	Dropped element	Product 12 but sum 8; checks only the constant.
D	Inverted operation	Gives $-7x$, the wrong sign on the middle term.

10. Answer A Concept 7

A teacher writing a quiz ~~on a laptop with a cracked screen~~ defines $f(x) = 2x + 1$ and $g(x) = x^2$. ~~The quiz has 10 questions.~~ What is $f(g(3))$?

Target word: value of a composition **Decides:** Apply g first: $g(3) = 9$.

A	Correct	$g(3) = 9$; $f(9) = 19$.
B	Adjacent category	$f(3) + g(3)$ adds the outputs.
C	Confusion set	$f(3) \times g(3)$ multiplies instead of composing.
D	Inverted operation	$g(f(3)) = 7^2$ applies the functions in the wrong order.

11. Answer D Concept 7 • transfer

$h(x) = (x + 4)^2$ is written as $f(g(x))$ with $f(x) = x^2$. What is $g(x)$?

Target word: decompose a composition **Decides:** g is the inner step that f squares.

A	Confusion set	$f(g(x))$ would be $(x^2 + 4)^2$.
B	Inverted operation	Gives $(x - 4)^2$, the wrong shift.
C	Dropped element	Leaves no work for f ; $f(g(x))$ would be $(x + 4)^4$.
D	Correct	$f(g(x)) = (x + 4)^2$.

12. Answer A Concept 8

On a Tuesday quiz ~~with 12 items~~, a student writes $2(x + 5) = 2x + 5$ on three problems. ~~The student's other answers were neat and on time.~~ What should the teacher do next?

Target word: response to a patterned error **Decides:** The error is consistent: 2 multiplies only the first term.

A	Correct	Targets the specific misconception with a check the student can see.
B	Confusion set	The facts are fine; the error is about what the 2 multiplies.
C	Adjacent category	Practice without diagnosis can reinforce the same error.
D	Absolute language	One patterned error is a teachable misconception, not a verdict.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	The different jobs a variable does	A letter can be a specific unknown ($3x + 5 = 11$), a generalized number in an identity ($a + b = b + a$), or a quantity that varies ($y = 2x + 3$).	Teaching that a letter always stands for one hidden number.	A set of equations and “what role does the letter play?”
2	Using structure before procedure	Seeing an expression’s structure, such as $(\text{something})^2 = 16$, lets you solve directly: $\text{something} = \pm 4$.	Expanding everything first, or keeping only the positive root.	An expression with a repeated chunk or a perfect square.
3	Arithmetic vs. geometric sequences	Arithmetic: $a_n = a_1 + (n - 1)d$, adding a constant; geometric: $a_n = a_1 \cdot r^{(n - 1)}$, multiplying by a constant.	Using n instead of $n - 1$, or treating repeated addition as multiplication.	“Each term is ___ more / ___ times the one before.”
4	Average rate of change (covariation)	Average rate of change on $[a, b] = (f(b) - f(a)) / (b - a)$; it links the change in output to the change in input.	Reporting only the change in output, or dividing a single value by b .	“Average speed” or “average rate” between two input values.
5	Historical stages of algebra	Algebra moved from rhetorical (all words) to syncopated (some abbreviations) to symbolic notation; many cultures, including 9th-century Baghdad, solved equations geometrically.	Believing algebra only counts once letters and symbols appear.	A historical problem written entirely in words.
6	Area models for multiplying and factoring	In an area model the dimensions are the factors and the total area is the product: $(x + 3)(x + 2) = x^2 + 5x + 6$.	Leaving out the middle rectangles, as in $(a + b)^2 = a^2 + b^2$.	Algebra tiles, a rectangle diagram, or “find the dimensions.”
7	Composition order	$f(g(x))$ applies g first, then f ; composition is generally not commutative and is not multiplication.	Evaluating left to right, or multiplying $f(x)$ by $g(x)$.	Notation like $f(g(3))$ or “written as a composition.”
8	Diagnose the error before reteaching	A wrong answer is evidence of a specific misconception; name it, then use a targeted representation or counterexample to address it.	Assigning more of the same practice without finding the cause.	A sample of student work and “what should the teacher do next?”

Course scope (public description): Algebra for Secondary Mathematics Teaching offers an in-depth exploration of advanced algebraic concepts and instructional methodologies tailored for secondary education. The course emphasizes task-based learning, enabling participants to engage deeply with algebra as an extension of number theory, operations, and abstract reasoning. Students will explore key algebraic ideas, including equivalence, patterns of change through covariation, and the integration of multiple representations—such as tables, graphs, equations, and geometric models. The course also examines the historical evolution of algebra and incorporates diverse cultural perspectives. Core topics include variables, functions (including exponential, logarithmic, polynomial, rational, and quadratic functions), and their applications. This course is designed to enhance educators' ability to teach complex algebraic concepts effectively and adapt instructional strategies to meet diverse student needs.